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Course Name: Data structure

Course code: CSA 0302

Q:1 Find the bugs , including the missing lines in the following snippet

Your Fixed Code

```
1 #include<stdio.h>
2 int linearsearch(int arr[],int n,int key){
3     for(int i=0;i<n;i++){
4         if(arr[i]==key){
5             return i;
6         }
7     }
8     return -1;
9 }
10 int main(){
11     int arr[]={10,20,30,40,50};
12     int key=30;
13     int result=linearsearch(arr,5,key);
14
15     if(result!=-1)
16         printf("found at index%d",result);
17     else
18         printf("Not found");
19     return 0;
20 }
```

Submitted

Test Input

30

Output

found at index2

Pending manual review

Q:2 Buggy code in an array

The screenshot shows a programming editor interface with three main panels: QUESTIONS, PROGRAM EDITOR, and INPUT/OUTPUT. The QUESTIONS panel on the left shows a question titled 'CHARACTER' with a description: 'Write a program that takes a character as input and determines whether it is a vowel or a consonant.' The PROGRAM EDITOR panel in the center contains a C program with several bugs. The INPUT panel on the right shows a single character 'I'. The OUTPUT panel on the right is empty. A Snipping Tool notification is visible in the bottom right corner, stating 'Screenshot copied to clipboard' and 'Automatically saved to screenshots folder.' The program code in the editor is as follows:

```
1#include <stdio.h>
2int main()
3{
4    char ch;
5    printf("%c", &ch);
6    if(ch=='a' || ch=='e' || ch=='i' || ch=='o' || ch=='u' || ch=='A' || ch=='E' || ch=='I' || ch=='O' || ch=='U')
7        printf("Vowels");
8    else
9        printf("Consonant");
10    return 0;
11}
```

Q:3 Identify an errors in the following snippet

```

1 #include<stdio.h>
2 int binarysearch(int arr[],int n,int key){
3     int low=0,high=n-1;
4     while(low<high){
5         int mid=(low+high)/2;
6         if (arr[mid]==key)
7             return mid;
8         else if(arr[mid]<key)
9             low=mid+1;
10        else
11            high=mid-1;
12    }
13    return -1;
14 }
15 int main(){
16     int arr[]={10,20,30,40,50,6,70,};
17     int n=7;
18     int key;
19
20     printf("Enter the element to search");
21     scanf("%d",&key);
22
23     int result=binarysearch(arr,n,key);
24
25     if(result!=-1)
26         printf("element found at index%d\n",result);
27     else
28         printf("element not found\n");

```

40

Output

```

Enter the element to search40
found at index3

```

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